GHRaisoni International Skill Tech University, Pune

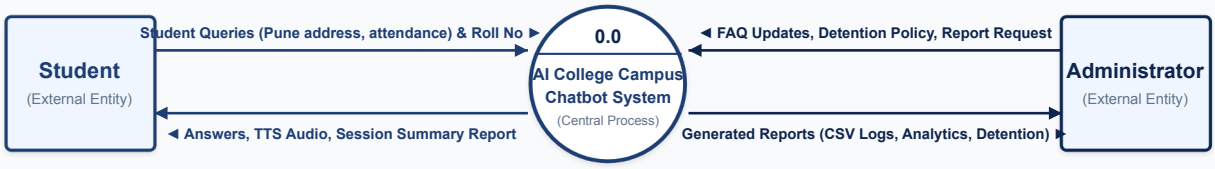
AI CHATBOT FOR COLLEGE

FACULTY REVISED SYSTEM ARCHITECTURE & DFD SPECIFICATION

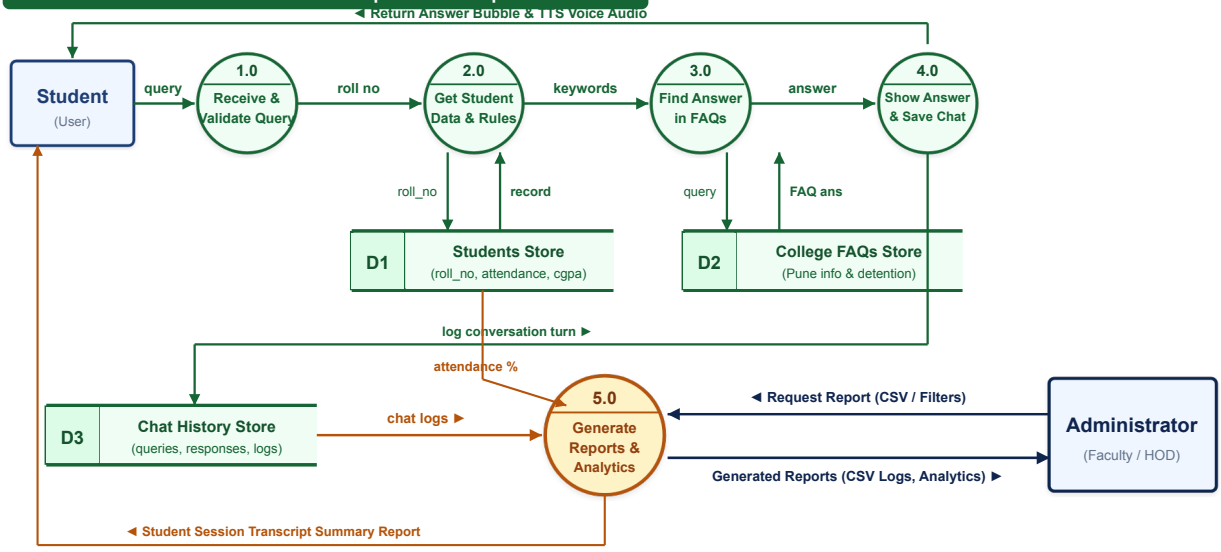
✓ Context-Level DFD (Level 0) Included ✓ Standard Open-Ended Data Stores ✓ Circle Process Bubbles (DeMarco/Gane-Sarson)

Data Flow Diagram (Level 0 Context & Level 1 Process Breakdown with Report Generation)

DFD Level 0: Context Diagram



DFD Level 1: First-Level Process Decomposition & Report Generation



Standard DFD Notation & Symbol Conventions (Gane-Sarson & DeMarco)

Symbol Shape	DFD Element	Standard Academic Definition	Implementation in Our Project
▭	External Entity	Origin/destination of data outside boundary (Square)	Student (User) & Administrator (Faculty/HOD)
○	Process (Bubble)	Transforms input data into output data (Circle with ID) Calculates, retrieves, formats, or generates reports	1.0 to 4.0 Query Handlers & 5.0 Generate Reports Query answering & administrative CSV/analytics reporting
D ═	Data Store *(Open-Ended)*	Repository of data at rest (Open parallel lines) **MANDATORY RULE: Right side must be open; NEVER closed**	D1 Students, D2 College FAQs, D3 Chat History Persistent SQLite tables feeding search & report generation
→	Data Flow	Directed solid arrow showing data path & payload Transfers packets between entities, processes, and stores	query, record, FAQ answer, generated reports CSV Directional vectors with explicit noun-phrase labels

Fig: Data Flow Diagram (DFD) — Top: Level 0 Context Diagram showing system boundary with Student & Administrator; Middle: Level 1 Process Decomposition (1.0 to 4.0 Query Handlers & 5.0 Generate Reports & Analytics) with standard open-ended parallel line data stores (D1 Students, D2 FAQs, D3 Chat History); Bottom: Standard DFD notation and symbols reference table.